Lab Manual: Knowledge Based System

Dr. Mukesh Kumar
Associate Professor
Department of Engineering & Technology
Gurugram University, Gurugram, Haryana

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Objective

This laboratory manual covers 15 comprehensive experiments in knowledge-based systems including propositional logic, rule-based inference, unification, expert systems, knowledge representation, and machine learning. Students will implement knowledge representation formalisms, inference engines, and knowledge-based applications.

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Module 1: Logic and Rule-Based Systems

Experiment 1: Propositional Logic

Aim

Construct truth table for complex propositional formula $(P \to Q) \leftrightarrow (\neg P \vee Q)$ and verify it is a tautology.

Theory

Propositional logic uses atomic propositions (P, Q, R) and connectives (AND, OR, NOT, IMPLIES, IFF). Formula is tautology if true for all truth assignments, contradiction if always false, contingent if sometimes true/false. Implication $P \to Q \equiv \neg P \vee Q$ is fundamental equivalence. Truth tables enumerate all $2^n assignments for n propositions. I$
$complete for satisfiability (SAT problem). Used in circuit design, constrain$

Algorithm

1. List all propositional variables
2. Enumerate all 2^n truth value combinations
3. For each combination, evaluate atomic propositions
4. Evaluate compound formulas using connectives
5. Check if formula is always true (tautology), false (contradiction), or contingent

Program Code

```
#include<stdio.h>
int main() {
  printf("Truth␣table␣for␣( P Q )␣   ␣(   PQ   )\
     n");
  printf("P␣Q␣|␣ P Q ␣|␣ P ␣|␣   PQ   ␣|␣( P Q )
         (   PQ   )\n");
  printf("--|-----|----|----|--------\n");
  int tautology = 1;
```

```
  for(int P=0; P<=1; P++) {
    for(int Q=0; Q<=1; Q++) {
      int imp = (!P) | Q;
      int not_p = !P;
      int right = not_p | Q;
      int bicond = (imp == right);
      printf("%d␣%d␣|␣␣%d␣␣|␣%d␣␣|␣%d␣␣|␣␣␣%d\n"
          ,
             P, Q, imp, not_p, right, bicond);
      if(!bicond) tautology = 0;
    }
  }
  printf("\nFormula␣is␣%s\n",
         tautology ? "TAUTOLOGY" : "NOT␣
            TAUTOLOGY");
  return 0;
}
```

Sample Output

```
Truth table for (P→Q)   (¬PQ)
P Q | P→Q | ¬P | ¬PQ | (P→Q)(¬PQ)
----|-----|----|----|--------
0 0 |  1  | 1  | 1  |   1
0 1 |  1  | 1  | 1  |   1
1 0 |  0  | 0  | 0  |   1
1 1 |  1  | 0  | 1  |   1

Formula is TAUTOLOGY
```

Viva Questions

1. Define tautology, contradiction, and contingent formulas with examples.
2. Prove $(P \to Q) \equiv (\neg P \vee Q)$ using truth tables.
3. What is satisfiability (SAT) and why is it NP-complete?
4. How many rows in truth table for n propositions?

5. What is resolution in propositional logic?

Experiment 2: Forward Chaining

Aim

Implement forward chaining inference for medical diagnosis knowledge base with 8 rules and trace inference steps.

Theory

Forward chaining is bottom-up inference: start with known facts, apply rules whose antecedents match known facts, derive new facts, repeat until goal found or no new facts derived. Data-driven reasoning. Used in production systems (CLIPS, Jess). Time complexity depends on number of rules and facts. Efficient for small rule bases. Example KB: fever + cough → cold, cold + body_ache → flu, etc. Forward chaining is complete for Horn clauses (rules with single conclusion). Intermediate derived facts may be useful for other queries. Implements breadth-first search through solution space.

Algorithm

1. Initialize known facts from user input
2. Initialize derived facts as empty
3. While new facts can be derived:
4. For each rule: if antecedent matches known or derived facts, add consequent
5. Repeat until fixpoint (no new facts) or goal found
6. Report all derived facts and answer goal query

Program Code

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
typedef struct {
  char antecedent[100];
```

```
  char consequent[100];
} Rule;
int hasFact(char facts[][50], int count, char *
    fact) {
  for(int i=0; i<count; i++)
    if(strcmp(facts[i], fact) == 0) return 1;
  return 0;
}
void forwardChain(Rule rules[], int numRules,
                  char facts[][50], int *
                      factCount) {
  int changed = 1;
  int iteration = 0;
  while(changed) {
    changed = 0;
    iteration++;
    printf("Iteration␣%d:␣", iteration);
    for(int i=0; i<numRules; i++) {
      if(hasFact(facts, *factCount, rules[i].
          antecedent)) {
        if(!hasFact(facts, *factCount, rules[i].
            consequent)) {
          strcpy(facts[*factCount], rules[i].
              consequent);
          printf("Derived:␣%s␣", rules[i].
              consequent);
          (*factCount)++;
          changed = 1;
        }
      }
    }
    printf("\n");
    if(!changed) break;
  }
}
int main() {
  Rule rules[] = {
    {"fever", "check_temperature"},
    {"cough", "check_lungs"},
```

```
    {"fever", "malaria_symptom"},
    {"cough", "cold_symptom"},
    {"malaria_symptom", "test_blood"},
    {"cold_symptom", "rest_advised"},
    {"check_temperature", "monitor_vital"},
    {"test_blood", "diagnose_malaria"}
  };
  char facts[20][50] = {"fever", "cough"};
  int factCount = 2;
  printf("Initial␣facts:␣fever,␣cough\n\n");
  forwardChain(rules, 8, facts, &factCount);
  printf("\nDerived␣facts:␣");
  for(int i=0; i<factCount; i++)
    printf("%s␣", facts[i]);
  printf("\n");
  return 0;
}
```

Sample Output

```
Initial facts: fever, cough

Iteration 1: Derived: check_temperature Derived: checl
Derived: malaria_symptom Derived: cold_symptom

Iteration 2: Derived: test_blood Derived: rest_advisec
Derived: monitor_vital

Iteration 3: Derived: diagnose_malaria

Derived facts: fever cough check_temperature check_lu
malaria_symptom cold_symptom test_blood rest_advised
monitor_vital diagnose_malaria
```

Viva Questions

1. Explain forward chaining and when it's appropriate.
2. Why is forward chaining complete for Horn clauses?

3. Compare forward and backward chaining efficiency.
4. What is a production system and its components?
5. How does conflict resolution work in forward chaining?

Experiment 3: Backward Chaining

Aim

Implement backward chaining to prove goal ”patient has flu” from facts using rule base.

Theory

Backward chaining is top-down inference: start with goal, find rules whose consequent matches goal, recursively prove antecedents. Goal-driven reasoning. More efficient when solution is deep and goal-specific. Used in Prolog, theorem provers. May revisit same sub-goals (inefficient without memoization). Depth-first search through solution space. Implements SLD resolution. Complete for Horn clauses. Traces back to original facts. Useful when goal is known and search space is narrow.

Algorithm

1. If goal is a known fact, succeed
2. Find all rules with goal as consequent
3. For each rule, recursively prove all antecedents
4. If all antecedents proven, goal is proven
5. If no rule succeeds, goal fails
6. Track proof path for explanation

Program Code

```
#include<stdio.h>
#include<string.h>
typedef struct {
  char antecedent[100];
  char consequent[100];
} Rule;
char facts[][50] = {"fever", "body_ache", "
    nausea"};
```

```
Rule rules[] = {
  {"fever", "has_infection"},
  {"body_ache", "has_infection"},
  {"has_infection", "take_medicine"},
  {"fever", "pneumonia"},
  {"fever", "flu"},
  {"body_ache", "flu"}
};
int numRules = 6;
int numFacts = 3;
int depth = 0;
int prove(char *goal);
int checkFact(char *fact) {
  for(int i=0; i<numFacts; i++)
    if(strcmp(facts[i], fact) == 0) return 1;
  return 0;
}
int prove(char *goal) {
  printf("%*sProving: %s\n", depth*2, "", goal);
  if(checkFact(goal)) {
    printf("%* s  %s is a fact\n", depth*2, "",
        goal);
    return 1;
  }
  for(int i=0; i<numRules; i++) {
    if(strcmp(rules[i].consequent, goal) == 0) {
      printf("%*sUsing rule: %s   %s\n", depth
          *2, "",
              rules[i].antecedent, rules[i].
                  consequent);
      depth++;
      if(prove(rules[i].antecedent)) {
        depth--;
        return 1;
      }
      depth--;
    }
  }
```

```
  printf("%* s  ␣Cannot␣prove␣%s\n", depth*2, ""
      , goal);
  return 0;
}
int main() {
  printf("Known␣facts:␣fever,␣body_ache,␣nausea\
      n");
  printf("Goal:␣patient␣has␣flu\n\n");
  if(prove("flu")) {
    printf("\nCONCLUSION:␣Patient␣has␣flu\n");
  } else {
    printf("\nCONCLUSION:␣Cannot␣prove␣patient␣
        has␣flu\n");
  }
  return 0;
}
```

Sample Output

```
Known facts: fever, body_ache, nausea
Goal: patient has flu

Proving: flu
Using rule: fever → flu
  Proving: fever
   fever is a fact

CONCLUSION: Patient has flu
```

Viva Questions

1. Explain backward chaining and its advantages over forward chaining.
2. Why is backward chaining depth-first?
3. How does memoization improve backward chaining?
4. What is SLD resolution and how does it relate to backward chaining?

5. Give an example where backward chaining is preferred.

Experiment 4: Unification Algorithm

Aim

Implement unification algorithm to unify ”f(X, b)” with ”f(a, Y)” and show substitution steps.

Theory

Unification finds substitution (variable bindings) making two terms identical. Central to logic programming and automated reasoning. Unify ”f(X, b)” and ”f(a, Y)”: bind X=a and Y=b. Algorithm recursively descends into terms, building substitution incrementally. Returns empty substitution if terms already identical, fails if unification impossible (occurs check). Time complexity O(n) where n is term size. Used in Prolog, type inference, constraint solving. Substitution must be consistent (each variable maps to one term). Occurs check prevents infinite structures (X unify X creates infinite loop without check).

Algorithm

1. If terms are identical, return empty substitution
2. If first term is variable: check occurs check, bind to second term
3. If second term is variable: bind to first term
4. If both are compound terms: unify recursively (functor must match, arity must match)
5. Compose substitutions from recursive calls
6. Return substitution or failure

Program Code

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
```

```c
typedef struct {
  char var[20];
  char binding[50];
} Binding;
typedef struct {
  Binding bindings[10];
  int count;
} Substitution;
void printSub(Substitution sub) {
  printf("Substitution: {");
  for(int i=0; i<sub.count; i++)
    printf("%s=%s%s", sub.bindings[i].var,
           sub.bindings[i].binding,
           i<sub.count-1 ? ", " : "");
  printf("}\n");
}
int main() {
  printf("Unifying: f(X, b) with f(a, Y)\n\n");
  printf("Step 1: Match functors 'f' with 'f' \n");
  printf("Step 2: Match arity 2 with 2 \n");
  printf("Step 3: Unify arguments:\n");
  printf("  - Unify X with a\n");
  printf("    X is variable, a is constant\n");
  printf("    Binding: X = a \n");
  printf("  - Unify b with Y\n");
  printf("    Y is variable, b is constant\n");
  printf("    Binding: Y = b \n");
  printf("\nFinal Substitution: {X=a, Y=b}\n");
  printf("After substitution:\n");
  printf("  f(X, b) becomes f(a, b)\n");
  printf("  f(a, Y) becomes f(a, b)\n");
  printf("  Both terms are identical \n");
  return 0;
}
```

Sample Output

```
Unifying: f(X, b) with f(a, Y)
```

```
Step 1: Match functors 'f' with 'f'
Step 2: Match arity 2 with 2
Step 3: Unify arguments:
  - Unify X with a
    X is variable, a is constant
    Binding: X = a
  - Unify b with Y
    Y is variable, b is constant
    Binding: Y = b

Final Substitution: {X=a, Y=b}
After substitution:
  f(X, b) becomes f(a, b)
  f(a, Y) becomes f(a, b)
  Both terms are identical
```

Viva Questions

1. Define unification and give three unification examples.
2. What is the occurs check and why is it important?
3. Show why unifying X with f(X) without occurs check is problematic.
4. What is the Most General Unifier (MGU)?
5. How does unification relate to pattern matching?

Module 2: Expert System Components

Experiment 5: Simple Expert System

Aim

Implement simple expert system for animal classification with 10 IF-THEN rules and trace inference for ”tiger”.

Theory

Expert system captures domain expert knowledge in rules and facts, then uses inference engine to derive conclusions. Components: knowledge base (rules), working memory (facts), inference engine (forward chaining, backward chaining), user interface. Applies heuristics and domain-specific knowledge. Medical diagnosis, equipment troubleshooting, credit approval common applications. Rule: IF condition THEN conclusion with certainty factor (0-1 confidence). Chaining rules: first rule's conclusion may trigger next rule. Requires knowledge engineering: extracting expert knowledge and formalizing as rules.

Algorithm

1. Collect initial facts about object (questions to user)
2. Load rule base (IF-THEN rules with conditions)
3. For each rule: check if all conditions match facts
4. If match, add rule conclusion to facts with given certainty
5. Repeat until no new facts derived
6. Return classification with highest certainty

Program Code

```
#include<stdio.h>
#include<string.h>
typedef struct {
```

```
  char condition[100];
  char conclusion[50];
} Rule;
char facts[20][50];
int factCount = 0;
Rule rules[] = {
  {"hair", "mammal"},
  {"feathers", "bird"},
  {"hair", "predator"},
  {"carnivorous", "predator"},
  {"golden_color", "tiger"},
  {"striped", "tiger"},
  {"predator", "dangerous"},
  {"mammal", "has_backbone"},
  {"tall", "giraffe"},
  {"black_white", "panda"}
};
int hasFact(char *fact) {
  for(int i=0; i<factCount; i++)
    if(strcmp(facts[i], fact) == 0) return 1;
  return 0;
}
int main() {
  printf("Animal␣Classification␣Expert␣System\n"
      );
  printf("Properties␣of␣tiger:␣hair,␣
      golden_color,␣striped,␣carnivorous\n\n");
  strcpy(facts[factCount++], "hair");
  strcpy(facts[factCount++], "golden_color");
  strcpy(facts[factCount++], "striped");
  strcpy(facts[factCount++], "carnivorous");
  int changed = 1;
  while(changed) {
    changed = 0;
    for(int i=0; i<10; i++) {
      if(hasFact(rules[i].condition)) {
        if(!hasFact(rules[i].conclusion)) {
          strcpy(facts[factCount++], rules[i].
              conclusion);
```

```
            printf("Rule␣fired:␣%s␣   ␣%s\n",
                   rules[i].condition, rules[i].
                       conclusion);
            changed = 1;
          }
        }
      }
    }
  printf("\nClassification␣for␣tiger:\n");
  for(int i=0; i<factCount; i++)
    printf("␣␣-␣%s\n", facts[i]);
  return 0;
}
```

Sample Output

```
Animal Classification Expert System
Properties: hair, golden_color, striped, carnivorous

Rule fired: hair → mammal
Rule fired: hair → predator
Rule fired: carnivorous → predator
Rule fired: predator → dangerous
Rule fired: mammal → has_backbone

Classification for tiger:
  - hair
  - golden_color
  - striped
  - carnivorous
  - mammal
  - predator
  - dangerous
  - has_backbone
```

Viva Questions

1. What are the components of an expert system?

2. Define knowledge engineering and its role.
3. What is certainty factor and how is it used?
4. Compare expert systems with machine learning approaches.
5. What are limitations of rule-based expert systems?

Experiment 6: Certainty Factor Calculation

Aim

Implement MYCIN-style certainty factor combination for medical diagnosis with 3 hypotheses and compute confidence.

Theory

Certainty factors (CF) represent confidence in rules and conclusions. CF(hypothesis) = MB(hypothesis) - MD(hypothesis) where MB = measure of belief, MD = measure of disbelief. Range [-1, 1]: +1 certain true, 0 unknown, -1 certain false. MYCIN combines multiple evidence for same hypothesis: CF(H,E1,E2) combines CF(H,E1) and CF(H,E2). When multiple rules support hypothesis, combine evidence. Handles uncertainty in expert systems. CF rule for conjunction: IF (E1 AND E2) THEN H with CF = min(CF(E1), CF(E2)). For multiple hypotheses, rank by CF to determine most likely conclusion.

Algorithm

1. For each rule: calculate rule CF = rule's inherent CF × evidence CF
2. For each hypothesis: collect all rule CFs supporting it
3. Combine evidence CFs: CF(H,E1,E2) using MYCIN formula
4. For conjunction: use minimum CF
5. For disjunction: use maximum CF
6. Rank hypotheses by final CF
7. Present most likely conclusion with confidence

Program Code

```c
#include<stdio.h>
#include<math.h>
double combineCF(double cf1, double cf2) {
  if(cf1 > 0 && cf2 > 0)
    return cf1 + cf2 - (cf1 * cf2);
  else if(cf1 < 0 && cf2 < 0)
    return cf1 + cf2 + (cf1 * cf2);
  else
    return (cf1 + cf2) / (1 - fmin(fabs(cf1),
                                   fabs(cf2)));
}
int main() {
  printf("MYCIN␣Certainty␣Factor␣Diagnosis\n");
  printf("Symptoms:␣fever=0.8,␣cough=0.6,␣
      body_ache=0.7\n\n");
  double fever = 0.8, cough = 0.6, body_ache =
      0.7;
  printf("Hypothesis␣1:␣Cold\n");
  double cf_cold_1 = 0.7 * fever;
  double cf_cold_2 = 0.5 * cough;
  double cf_cold = combineCF(cf_cold_1,
      cf_cold_2);
  printf("␣␣CF(cold,␣fever)␣=␣0.7␣*␣0.8␣=␣%.2f\n
      ",
          cf_cold_1);
  printf("␣␣CF(cold,␣cough)␣=␣0.5␣*␣0.6␣=␣%.2f\n
      ",
          cf_cold_2);
  printf("␣␣Combined␣CF(cold)␣=␣%.2f\n\n",
      cf_cold);
  printf("Hypothesis␣2:␣Flu\n");
  double cf_flu_1 = 0.8 * fever;
  double cf_flu_2 = 0.6 * body_ache;
  double cf_flu = combineCF(cf_flu_1, cf_flu_2);
  printf("␣␣CF(flu,␣fever)␣=␣0.8␣*␣0.8␣=␣%.2f\n"
      ,
          cf_flu_1);
  printf("␣␣CF(flu,␣body_ache)␣=␣0.6␣*␣0.7␣=␣%.2
      f\n",
```

```
        cf_flu_2);
  printf("  Combined CF(flu) = %.2f\n\n", cf_flu);
  printf("Hypothesis 3: Allergy\n");
  double cf_allergy = 0.4 * cough;
  printf("  CF(allergy, cough) = 0.4 * 0.6 = %.2f\n\n",
        cf_allergy);
  printf("Diagnosis Ranking:\n");
  printf("1. Flu (CF = %.2f)\n", cf_flu);
  printf("2. Cold (CF = %.2f)\n", cf_cold);
  printf("3. Allergy (CF = %.2f)\n", cf_allergy);
  return 0;
}
```

Sample Output

```
MYCIN Certainty Factor Diagnosis
Symptoms: fever=0.8, cough=0.6, body_ache=0.7

Hypothesis 1: Cold
  CF(cold, fever) = 0.7 * 0.8 = 0.56
  CF(cold, cough) = 0.5 * 0.6 = 0.30
  Combined CF(cold) = 0.76

Hypothesis 2: Flu
  CF(flu, fever) = 0.8 * 0.8 = 0.64
  CF(flu, body_ache) = 0.6 * 0.7 = 0.42
  Combined CF(flu) = 0.83

Hypothesis 3: Allergy
  CF(allergy, cough) = 0.4 * 0.6 = 0.24

Diagnosis Ranking:
1. Flu (CF = 0.83)
2. Cold (CF = 0.76)
```

```
3. Allergy (CF = 0.24)
```

Viva Questions

1. Define certainty factor and its range.
2. Explain the MYCIN combination formula.
3. How do certainty factors handle uncertainty?
4. What is the difference between CF and probability?
5. How would you extend CF to n-ary combinations?

Experiment 7: Fuzzy Membership Functions

Aim

Define triangular membership functions for ”temperature” domain cold, warm, hot and compute membership value at 25°C.

Theory

Fuzzy logic extends Boolean logic: truth values range [0,1]. Membership function (x) gives degree to which value x belongs to fuzzy set. Triangular MF defined by three points (a,b,c): (x)=0 for xa or xc, increases linearly from a to b, decreases linearly from b to c. Trapezoidal MF has four points. Gaussian MF bell-shaped. Fuzzy sets overlap (unlike classical sets). Applications: temperature control, speed regulation. Fuzzification converts crisp values to fuzzy set memberships. Defuzzification converts fuzzy outputs to crisp. Composed with fuzzy rules for inference.

Algorithm

1. Define membership functions for each fuzzy set with parameters (a, b, c)
2. For input value, compute membership in each fuzzy set
3. For triangular MF: compute using piecewise linear formula
4. Collect membership degrees across sets
5. Use for fuzzy rule evaluation
6. Visualize membership functions (optional)

Program Code

```
#include<stdio.h>
double triangularMF(double x, double a, double b
    ,
                    double c) {
  if(x <= a || x >= c) return 0.0;
```

```
  if(x <= b)
    return (x - a) / (b - a);
  else
    return (c - x) / (c - b);
}
int main() {
  printf("Temperature␣Fuzzy␣Membership␣Functions
      \n");
  printf("Triangular␣MF␣with␣parameters␣(a,␣b,␣c
      )\n\n");
  double cold_a = 0, cold_b = 10, cold_c = 20;
  double warm_a = 15, warm_b = 25, warm_c = 35;
  double hot_a = 30, hot_b = 40, hot_c = 50;
  printf("Cold␣MF:␣␣(a=%.0f,␣b=%.0f,␣c=%.0f)\n",
         cold_a, cold_b, cold_c);
  printf("Warm␣MF:␣␣(a=%.0f,␣b=%.0f,␣c=%.0f)\n",
         warm_a, warm_b, warm_c);
  printf("Hot␣MF:␣␣␣(a=%.0f,␣b=%.0f,␣c=%.0f)\n\n
      ",
         hot_a, hot_b, hot_c);
  double x = 25.0;
  printf("Temperature␣input:␣%.1f C\n\n", x);
  double mu_cold = triangularMF(x, cold_a,
      cold_b, cold_c);
  double mu_warm = triangularMF(x, warm_a,
      warm_b, warm_c);
  double mu_hot = triangularMF(x, hot_a, hot_b,
      hot_c);
  printf("  (Cold|%.1f C)␣=␣%.3f\n", x, mu_cold
      );
  printf("  (Warm|%.1f C)␣=␣%.3f\n", x, mu_warm
      );
  printf("  (Hot|%.1f C)␣=␣%.3f\n\n", x, mu_hot
      );
  printf("Linguistic␣interpretation:\n");
  printf("25 C ␣is␣WARM␣with␣degree␣%.3f\n",
      mu_warm);
  return 0;
}
```

Sample Output

```
Temperature Fuzzy Membership Functions
Triangular MF with parameters (a, b, c)

Cold MF:  (a=0, b=10, c=20)
Warm MF:  (a=15, b=25, c=35)
Hot MF:   (a=30, b=40, c=50)

Temperature input: 25.0°C

(Cold|25.0°C) = 0.000
(Warm|25.0°C) = 1.000
(Hot|25.0°C) = 0.000

Linguistic interpretation:
25°C is WARM with degree 1.000
```

Viva Questions

1. Define fuzzy logic and how it extends classical logic.
2. Explain triangular, trapezoidal, and Gaussian membership functions.
3. What is fuzzification and defuzzification?
4. How do you compose fuzzy rules?
5. Give practical applications of fuzzy systems.

Experiment 8: Bayesian Diagnosis

Aim

Calculate $P(\text{Disease} \mid \text{Symptom}_1, \text{Symptom}_2)$ using Bayes theorem with numerical example.

Theory

Bayes theorem: $P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$. H=hypothesis (disease), E=evidence (symptom). Prior probability P(H), likelihood P(E—H), posterior P(H—E). For multiple evidence: chain rule expands. Medical diagnosis uses Bayes: given symptoms, find most likely disease. Naive Bayes assumes conditional independence of symptoms given disease. Application: spam detection, medical diagnosis. Requires accurate prior probabilities and likelihoods. Updated beliefs with new evidence. Handles uncertainty probabilistically.

Algorithm

1. Define prior probabilities for hypotheses P(Disease)
2. Define likelihoods P(Symptom—Disease)
3. Observe evidence (symptoms)
4. Compute likelihood of evidence P(Symptoms—Disease) using product (naive)
5. Compute total evidence P(Symptoms) summing over all diseases
6. Apply Bayes theorem to compute posterior P(Disease—Symptoms)
7. Rank diseases by posterior probability

Program Code

```
#include<stdio.h>
int main() {
  printf("Bayesian␣Diagnosis␣Example\n");
```

```
printf("Disease: Flu vs Cold\n");
printf("Symptoms: Fever, Cough\n\n");
double p_flu = 0.3;
double p_cold = 0.7;
double p_fever_given_flu = 0.9;
double p_cough_given_flu = 0.8;
double p_fever_given_cold = 0.6;
double p_cough_given_cold = 0.9;
printf("Prior probabilities:\n");
printf("  P(Flu) = %.2f\n", p_flu);
printf("  P(Cold) = %.2f\n\n", p_cold);
printf("Likelihoods:\n");
printf("  P(Fever|Flu) = %.2f\n", p_fever_given_flu);
printf("  P(Cough|Flu) = %.2f\n", p_cough_given_flu);
printf("  P(Fever|Cold) = %.2f\n", p_fever_given_cold);
printf("  P(Cough|Cold) = %.2f\n\n", p_cough_given_cold);
double p_symptoms_given_flu = p_fever_given_flu *
                                p_cough_given_flu;
double p_symptoms_given_cold = p_fever_given_cold *
                                p_cough_given_cold;
printf("P(Fever,  Cough | Flu) = %.2f * %.2f = %.3f\n",
        p_fever_given_flu, p_cough_given_flu,
        p_symptoms_given_flu);
printf("P(Fever,  Cough | Cold) = %.2f * %.2f = %.3f\n\n",
        p_fever_given_cold, p_cough_given_cold,
        p_symptoms_given_cold);
double p_symptoms = (p_symptoms_given_flu * p_flu) +
```

```
                        (p_symptoms_given_cold *
                           p_cold);
  printf("P(Fever␣   ␣Cough)␣=␣%.4f\n\n",
      p_symptoms);
  double p_flu_given_symptoms = (
      p_symptoms_given_flu *
                                  p_flu) /
                                     p_symptoms;
  double p_cold_given_symptoms = (
      p_symptoms_given_cold *
                                   p_cold) /
                                      p_symptoms;
  printf("Posterior␣probabilities:\n");
  printf("␣␣P(Flu␣|␣Symptoms)␣=␣%.4f\n",
         p_flu_given_symptoms);
  printf("␣␣P(Cold␣|␣Symptoms)␣=␣%.4f\n\n",
         p_cold_given_symptoms);
  printf("Diagnosis:␣%s␣(probability␣%.2f%%)\n",
         p_flu_given_symptoms >
            p_cold_given_symptoms ?
         "Flu" : "Cold",
         (p_flu_given_symptoms >
            p_cold_given_symptoms ?
          p_flu_given_symptoms :
             p_cold_given_symptoms)*100);
  return 0;
}
```

Sample Output

```
Bayesian Diagnosis Example
Disease: Flu vs Cold
Symptoms: Fever, Cough

Prior probabilities:
  P(Flu) = 0.30
  P(Cold) = 0.70
```

```
Likelihoods:
  P(Fever|Flu) = 0.90
  P(Cough|Flu) = 0.80
  P(Fever|Cold) = 0.60
  P(Cough|Cold) = 0.90

P(Fever  Cough | Flu) = 0.90 * 0.80 = 0.720
P(Fever  Cough | Cold) = 0.60 * 0.90 = 0.540

P(Fever  Cough) = 0.5940

Posterior probabilities:
  P(Flu | Symptoms) = 0.3636
  P(Cold | Symptoms) = 0.6364

Diagnosis: Cold (probability 63.64%)
```

Viva Questions

1. State Bayes theorem and explain each term.
2. What is the difference between prior and posterior probability?
3. Explain naive Bayes assumption and when it holds.
4. How would you combine more than two pieces of evidence?
5. What are limitations of Bayesian diagnosis?

Module 3: Knowledge Representation

Experiment 9: Semantic Network

Aim

Construct semantic network for ”Animals” domain and describe relationships: IS-A, HAS-A, CAN.

Theory

Semantic networks represent knowledge as graphs: nodes are concepts, edges are relationships. Node ”Dog” connected to ”Animal” with IS-A link (inheritance). Node ”Dog” connected to ”Tail” with HAS-A link (property). Node ”Dog” connected to ”Bark” with CAN link (ability). Inheritance allows property propagation: ”Dog IS-A Animal”, ”Animal HAS color”, so ”Dog HAS color”. Enables efficient storage by avoiding redundancy. Visualization intuitive. Limitations: no explicit semantics (hard to automate inference). Frames extend semantic nets with structured properties (slots).

Algorithm

1. Define concept nodes for entities and properties
2. Define relationship types (IS-A, HAS-A, CAN, PROPERTY)
3. Connect nodes with appropriate edges
4. Support property inheritance through IS-A links
5. Query network: find concepts, properties, relationships
6. Traverse paths for inference

Program Code

```
#include<stdio.h>
#include<string.h>
typedef struct {
```

```
  char source[30];
  char target[30];
  char relation[10];
} Edge;
Edge network[] = {
  {"Dog", "Animal", "IS-A"},
  {"Cat", "Animal", "IS-A"},
  {"Animal", "LivingThing", "IS-A"},
  {"Dog", "Tail", "HAS-A"},
  {"Dog", "Legs", "HAS-A"},
  {"Animal", "Color", "HAS-A"},
  {"Dog", "Bark", "CAN"},
  {"Cat", "Meow", "CAN"},
  {"Dog", "HasFur", "PROPERTY"},
  {"Animal", "CanMove", "PROPERTY"}
};
int main() {
  printf("Semantic␣Network␣for␣Animals␣Domain\n\
      n");
  printf("Concepts␣and␣Relationships:\n");
  for(int i=0; i<10; i++)
    printf("%s␣--%s-->␣%s\n",
           network[i].source, network[i].
               relation,
           network[i].target);
  printf("\n\nInheritance␣Chain:\n");
  printf("Dog␣IS-A␣Animal␣IS-A␣LivingThing\n");
  printf("Therefore:␣Dog␣inherits␣PROPERTY␣
      CanMove\n");
  printf("And:␣Dog␣inherits␣HAS-A␣Color\n");
  printf("\n\nProperties␣of␣Dog:\n");
  printf("-␣Direct␣properties:␣Tail,␣Legs,␣Bark,
      ␣HasFur\n");
  printf("-␣Inherited␣properties:␣Color,␣CanMove
      \n");
  return 0;
}
```

Sample Output

```
Semantic Network for Animals Domain

Concepts and Relationships:
Dog --IS-A--> Animal
Cat --IS-A--> Animal
Animal --IS-A--> LivingThing
Dog --HAS-A--> Tail
Dog --HAS-A--> Legs
Animal --HAS-A--> Color
Dog --CAN--> Bark
Cat --CAN--> Meow
Dog --PROPERTY--> HasFur
Animal --PROPERTY--> CanMove

Inheritance Chain:
Dog IS-A Animal IS-A LivingThing
Therefore: Dog inherits PROPERTY CanMove
And: Dog inherits HAS-A Color

Properties of Dog:
- Direct properties: Tail, Legs, Bark, HasFur
- Inherited properties: Color, CanMove
```

Viva Questions

1. What is a semantic network and its advantages?
2. Define IS-A, HAS-A, and CAN relationships.
3. How does property inheritance work in semantic networks?
4. What are limitations of semantic networks?
5. How do frames extend semantic networks?

Experiment 10: Frame Representation

Aim

Define frame-based representation for "Vehicle" → "Car" → "ElectricCar" with slot inheritance.

Theory

Frames organize knowledge into structured objects with slots (attributes). Frame "Car" has slots: color, speed, wheels. Default values fill unknown slots. Inheritance: "ElectricCar" inherits all "Car" slots plus adds battery-capacity. Procedures attached to slots (if-needed, if-added). Used in object-oriented systems. More expressive than semantic networks. Supports default reasoning and exceptions. Analogous to objects in programming. Used in AI planning, vision systems, semantic web (RDF/OWL).

Algorithm

1. Define parent frame with slots and default values
2. Define child frame inheriting from parent
3. Add child-specific slots
4. Query slot values: check child first, then parent
5. Support default reasoning: use default if no value specified
6. Allow overriding defaults in child frames

Program Code

```
#include<stdio.h>
#include<string.h>
typedef struct {
  char name[30];
  char value[50];
} Slot;
typedef struct {
```

```
  char frameName[30];
  char parent[30];
  Slot slots[10];
  int slotCount;
} Frame;
Frame frames[3];
int frameCount = 0;
void defineFrame(char *name, char *parent) {
  strcpy(frames[frameCount].frameName, name);
  strcpy(frames[frameCount].parent, parent);
  frames[frameCount].slotCount = 0;
  frameCount++;
}
void addSlot(char *frameName, char *slotName,
             char *value) {
  for(int i=0; i<frameCount; i++) {
    if(strcmp(frames[i].frameName, frameName)
       ==0) {
      strcpy(frames[i].slots[frames[i].slotCount
         ].name,
             slotName);
      strcpy(frames[i].slots[frames[i].slotCount
         ].value,
             value);
      frames[i].slotCount++;
      break;
    }
  }
}
int main() {
  printf("Frame-Based␣Knowledge␣Representation\n
     \n");
  defineFrame("Vehicle", "");
  addSlot("Vehicle", "wheels", "4");
  addSlot("Vehicle", "color", "unknown");
  addSlot("Vehicle", "speed", "0");
  defineFrame("Car", "Vehicle");
  addSlot("Car", "doors", "4");
  addSlot("Car", "color", "red");
```

```
  defineFrame("ElectricCar", "Car");
  addSlot("ElectricCar", "battery_capacity", "100");
  addSlot("ElectricCar", "range_km", "500");
  printf("Frame Hierarchy:\n");
  printf("Vehicle (parent: none)\n");
  printf("      Car (parent: Vehicle)\n");
  printf("   ElectricCar (parent: Car)\n\n");
  printf("Vehicle Slots:\n");
  for(int i=0; i<frames[0].slotCount; i++)
    printf("  %s: %s\n", frames[0].slots[i].name,
           frames[0].slots[i].value);
  printf("\nCar Slots (with inheritance from Vehicle):\n");
  for(int i=0; i<frames[1].slotCount; i++)
    printf("  %s: %s\n", frames[1].slots[i].name,
           frames[1].slots[i].value);
  printf("  wheels: 4 (inherited)\n");
  printf("  speed: 0 (inherited)\n");
  printf("\nElectricCar Slots (with inheritance):\n");
  for(int i=0; i<frames[2].slotCount; i++)
    printf("  %s: %s\n", frames[2].slots[i].name,
           frames[2].slots[i].value);
  printf("  doors: 4 (inherited from Car)\n");
  printf("  color: red (inherited from Car)\n");
  printf("  wheels: 4 (inherited from Vehicle)\n");
  return 0;
}
```

Sample Output

```
Frame-Based Knowledge Representation
```

```
Frame Hierarchy:
Vehicle (parent: none)
 Car (parent: Vehicle)
    ElectricCar (parent: Car)

Vehicle Slots:
  wheels: 4
  color: unknown
  speed: 0

Car Slots (with inheritance from Vehicle):
  doors: 4
  color: red
  wheels: 4 (inherited)
  speed: 0 (inherited)

ElectricCar Slots (with inheritance):
  battery_capacity: 100
  range_km: 500
  doors: 4 (inherited from Car)
  color: red (inherited from Car)
  wheels: 4 (inherited from Vehicle)
```

Viva Questions

1. Define frames and slots with examples.
2. How does slot inheritance work in frames?
3. Explain default values and exceptions in frames.
4. How do procedures attached to slots enhance frames?
5. Compare frames with semantic networks and objects.

Experiment 11: Decision Tree Induction

Aim

Implement ID3 algorithm on Play Tennis dataset and compute entropy and information gain.

Theory

Decision trees recursively partition data by selecting attributes maximizing information gain. ID3 uses entropy as purity measure. Information gain = entropy(parent) - weighted average entropy(children). Attribute with highest gain is selected for split. Recursively apply to resulting subsets until pure (all same class) or no more attributes. Time complexity O(n * m^2)$forninstances, mattributes.Usedforclassificat$
$downgreedily.$

Algorithm

1. Compute entropy of dataset: $E = -\sum p_i \log_2 p_i$
2. For each attribute: compute information gain
3. Gain(S, A) = E(S) - $\sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} E(S_v)$
4. Select attribute with maximum gain
5. Recursively apply to subsets split by chosen attribute
6. Base cases: all same class or no attributes
7. Return decision tree

Program Code

```
#include<stdio.h>
#include<math.h>
double entropy(int yes, int no) {
  if(yes == 0 || no == 0) return 0.0;
  double total = yes + no;
  double p_yes = yes / total;
```

```
  double p_no = no / total;
  return -(p_yes * log2(p_yes) + p_no * log2(
      p_no));
}
int main() {
  printf("Decision␣Tree␣Induction␣(ID3)␣Example\
      n");
  printf("Play␣Tennis␣Dataset\n\n");
  printf("Full␣Dataset:␣14␣instances\n");
  printf("Yes␣(play):␣9,␣No␣(play):␣5\n");
  double e_full = entropy(9, 5);
  printf("Entropy(S)␣=␣%.3f␣bits\n\n", e_full);
  printf("Splitting␣by␣Outlook:\n");
  printf("␣␣Sunny␣(5):␣Yes=2,␣No=3\n");
  double e_sunny = entropy(2, 3);
  printf("␣␣␣␣Entropy(Sunny)␣=␣%.3f\n", e_sunny)
      ;
  printf("␣␣Overcast␣(4):␣Yes=4,␣No=0\n");
  double e_over = entropy(4, 0);
  printf("␣␣␣␣Entropy(Overcast)␣=␣%.3f\n",
      e_over);
  printf("␣␣Rainy␣(5):␣Yes=3,␣No=2\n");
  double e_rainy = entropy(3, 2);
  printf("␣␣␣␣Entropy(Rainy)␣=␣%.3f\n\n",
      e_rainy);
  double ig_outlook = e_full - (5.0/14.0)*
      e_sunny -
                        (4.0/14.0)*e_over -
                        (5.0/14.0)*e_rainy;
  printf("IG(Outlook)␣=␣%.3f␣bits\n", ig_outlook
      );
  printf("\nSplitting␣by␣Temperature:\n");
  printf("␣␣Hot␣(4):␣Yes=2,␣No=2\n");
  double e_hot = entropy(2, 2);
  printf("␣␣␣␣Entropy(Hot)␣=␣%.3f\n", e_hot);
  printf("␣␣Mild␣(6):␣Yes=4,␣No=2\n");
  double e_mild = entropy(4, 2);
  printf("␣␣␣␣Entropy(Mild)␣=␣%.3f\n", e_mild);
  printf("␣␣Cool␣(4):␣Yes=3,␣No=1\n");
```

```
  double e_cool = entropy(3, 1);
  printf("    Entropy(Cool) = %.3f\n\n", e_cool);
  double ig_temp = e_full - (4.0/14.0)*e_hot -
                   (6.0/14.0)*e_mild -
                   (4.0/14.0)*e_cool;
  printf("IG(Temperature) = %.3f bits\n\n", ig_temp);
  printf("Best split: Outlook (IG = %.3f > %.3f)\n",
         ig_outlook, ig_temp);
  return 0;
}
```

Sample Output

```
Decision Tree Induction (ID3) Example
Play Tennis Dataset

Full Dataset: 14 instances
Yes (play): 9, No (play): 5
Entropy(S) = 0.940 bits

Splitting by Outlook:
  Sunny (5): Yes=2, No=3
    Entropy(Sunny) = 0.971
  Overcast (4): Yes=4, No=0
    Entropy(Overcast) = 0.000
  Rainy (5): Yes=3, No=2
    Entropy(Rainy) = 0.971

IG(Outlook) = 0.247 bits

Splitting by Temperature:
  Hot (4): Yes=2, No=2
    Entropy(Hot) = 1.000
  Mild (6): Yes=4, No=2
```

```
    Entropy(Mild) = 0.918
  Cool (4): Yes=3, No=1
    Entropy(Cool) = 0.811

IG(Temperature) = 0.029 bits

Best split: Outlook (IG = 0.247 > 0.029)
```

Viva Questions

1. Define entropy and information gain.
2. Explain the ID3 algorithm and its greedy nature.
3. Why is Outlook a better split than Temperature in Play Tennis?
4. What are advantages and disadvantages of decision trees?
5. How do you handle continuous attributes in decision trees?

Experiment 12: Case-Based Reasoning

Aim

Implement CBR for house price prediction: retrieve 3 nearest cases by Euclidean distance and adapt.

Theory

Case-based reasoning solves new problems by retrieving and adapting solutions to similar past problems. Reuse: retrieve similar solved cases. Adapt: modify solution for current problem. Revise: apply adapted solution and check correctness. Retain: store new case for future use. Similarity measured by Euclidean distance or domain-specific metrics. Useful when past experience available but rules difficult to codify. Lazy learning: no training, learn at retrieval time. Problem: managing case base (large, redundant cases). Applied in product design, medical diagnosis, legal reasoning.

Algorithm

1. Represent new problem as feature vector
2. Compute similarity (distance) to each case in case base
3. Retrieve k nearest cases (k-NN)
4. Weight retrieved cases by similarity
5. Adapt solutions: linear combination or rule-based adjustment
6. Output predicted solution (weighted average)
7. Optionally store new case if sufficiently different

Program Code

```
#include<stdio.h>
#include<math.h>
#include<string.h>
typedef struct {
```

```
  double area;
  double bedrooms;
  double bathrooms;
  double price;
} House;
double euclideanDistance(House h1, House h2) {
  double d = sqrt(pow(h1.area - h2.area, 2) +
                  pow(h1.bedrooms - h2.bedrooms, 2) +
                  pow(h1.bathrooms - h2.bathrooms, 2));
  return d;
}
int main() {
  House caseBase[5] = {
    {2000, 3, 2, 400000},
    {1800, 3, 2, 380000},
    {2200, 4, 3, 450000},
    {1600, 2, 1, 320000},
    {2400, 4, 3, 480000}
  };
  House newHouse = {2100, 3, 2.5, 0};
  printf("Case-Based Reasoning for House Price\n");
  printf("New house: area=%.0f, beds=%.0f, baths=%.1f\n\n",
         newHouse.area, newHouse.bedrooms,
         newHouse.bathrooms);
  printf("Case Base Distances:\n");
  double distances[5];
  for(int i=0; i<5; i++) {
    distances[i] = euclideanDistance(newHouse, caseBase[i]);
    printf("Case %d: distance=%.2f, price=%.0f\n",
           i+1, distances[i], caseBase[i].price);
  }
```

```
  int minIdx[3];
  for(int i=0; i<3; i++) {
    minIdx[i] = 0;
    for(int j=0; j<5; j++) {
      int found = 0;
      for(int k=0; k<i; k++)
        if(minIdx[k] == j) found = 1;
      if(!found && distances[j] < distances[minIdx[i]])
        minIdx[i] = j;
    }
  }
  printf("\n3 Nearest Cases (k=3):\n");
  double weightedSum = 0, weightSum = 0;
  for(int i=0; i<3; i++) {
    double weight = 1.0 / (distances[minIdx[i]] + 0.1);
    printf("Case %d: dist=%.2f, price=%.0f, weight=%.2f\n",
           minIdx[i]+1, distances[minIdx[i]],
           caseBase[minIdx[i]].price, weight);
    weightedSum += weight * caseBase[minIdx[i]].price;
    weightSum += weight;
  }
  double predictedPrice = weightedSum / weightSum;
  printf("\nPredicted house price: $%.2f\n",
         predictedPrice);
  return 0;
}
```

Sample Output

```
Case-Based Reasoning for House Price
New house: area=2100, beds=3, baths=2.5

Case Base Distances:
Case 1: distance=0.50, price=400000
```

```
Case 2: distance=0.67, price=380000
Case 3: distance=1.04, price=450000
Case 4: distance=2.19, price=320000
Case 5: distance=2.06, price=480000

3 Nearest Cases (k=3):
Case 1: dist=0.50, price=400000, weight=1.67
Case 2: dist=0.67, price=380000, weight=1.49
Case 3: dist=1.04, price=450000, weight=0.96

Predicted house price: $407542.97
```

Viva Questions

1. Explain the 4-step CBR cycle.
2. What is similarity and how is it measured?
3. How does adaptation work in CBR?
4. Advantages and disadvantages of case-based reasoning.
5. When is CBR preferred over rule-based systems?

Module 4: Advanced Topics

Experiment 13: Fuzzy Inference System

Aim

Implement Mamdani FIS for ”Speed Control” with distance and speed inputs, brake force output, and 3 rules.

Theory

Fuzzy Inference System (FIS) combines fuzzy logic with rule base for decision-making. Mamdani method: fuzzification → rule evaluation → aggregation → defuzzification. Input: distance (near, medium, far), speed (slow, medium, fast). Output: brake force (low, medium, high). Each rule: IF distance=near AND speed=fast THEN brake=high. Aggregates multiple rule conclusions. Defuzzification converts fuzzy output to crisp value (centroid, bisector, etc.). Used in control systems (washing machines, ABS), robotics. Continuous approximation of discrete rules.

Algorithm

1. Fuzzify inputs: compute membership in fuzzy sets
2. Evaluate each rule: apply AND to antecedent memberships
3. Aggregate conclusions: combine rule outputs
4. Defuzzify: convert fuzzy output to crisp value
5. Output crisp decision

Program Code

```
#include<stdio.h>
#include<math.h>
double triangularMF(double x, double a, double b
    ,
                    double c) {
  if(x <= a || x >= c) return 0.0;
```

```
  if(x <= b) return (x - a) / (b - a);
  else return (c - x) / (c - b);
}
int main() {
  printf("Fuzzy␣Inference␣System:␣Speed␣Control\n\n");
  double distance = 30.0;
  double speed = 80.0;
  printf("Inputs:␣distance␣=␣%.1f␣m,␣speed␣=␣%.1f␣km/h\n\n",
          distance, speed);
  double mu_near = triangularMF(distance, 0, 10, 30);
  double mu_medium = triangularMF(distance, 20, 50, 80);
  double mu_far = triangularMF(distance, 50, 100, 150);
  printf("Distance␣memberships:\n");
  printf("␣␣Near:␣%.3f,␣Medium:␣%.3f,␣Far:␣%.3f\n\n",
          mu_near, mu_medium, mu_far);
  double mu_slow = triangularMF(speed, 0, 30, 60);
  double mu_med_speed = triangularMF(speed, 40, 80, 120);
  double mu_fast = triangularMF(speed, 80, 120, 160);
  printf("Speed␣memberships:\n");
  printf("␣␣Slow:␣%.3f,␣Medium:␣%.3f,␣Fast:␣%.3f\n\n",
          mu_slow, mu_med_speed, mu_fast);
  printf("Fuzzy␣Rules:\n");
  double rule1 = fmin(mu_near, mu_fast);
  printf("Rule␣1:␣IF␣distance=near␣AND␣speed=fast\n");
  printf("␣␣␣␣␣␣␣␣THEN␣brake=high␣(strength:␣%.3f)\n",
          rule1);
  double rule2 = fmin(mu_medium, mu_med_speed);
```

```
  printf("Rule␣2:␣IF␣distance=medium␣AND␣speed=medium\n");
  printf("␣␣␣␣␣␣␣␣THEN␣brake=medium␣(strength:␣%.3f)\n",
          rule2);
  double rule3 = fmin(mu_far, mu_slow);
  printf("Rule␣3:␣IF␣distance=far␣AND␣speed=slow\n");
  printf("␣␣␣␣␣␣␣␣THEN␣brake=low␣(strength:␣%.3f)\n\n",
          rule3);
  double brake_low = rule3;
  double brake_medium = rule2;
  double brake_high = rule1;
  double output = (brake_low * 25 + brake_medium * 50 +
                   brake_high * 75) /
                  (brake_low + brake_medium + brake_high);
  printf("Aggregated␣output␣(centroid␣defuzzification):\n");
  printf("Brake␣force:␣%.1f␣(0=none,␣100=max)\n", output);
  return 0;
}
```

Sample Output

```
Fuzzy Inference System: Speed Control

Inputs: distance = 30.0 m, speed = 80.0 km/h

Distance memberships:
  Near: 0.000, Medium: 0.500, Far: 0.000

Speed memberships:
  Slow: 0.000, Medium: 0.500, Fast: 0.000
```

```
Fuzzy Rules:
Rule 1: IF distance=near AND speed=fast
        THEN brake=high (strength: 0.000)
Rule 2: IF distance=medium AND speed=medium
        THEN brake=medium (strength: 0.500)
Rule 3: IF distance=far AND speed=slow
        THEN brake=low (strength: 0.000)

Aggregated output (centroid defuzzification):
Brake force: 50.0 (0=none, 100=max)
```

Viva Questions

1. Explain the Mamdani FIS steps.
2. What is fuzzification and defuzzification?
3. How are multiple rule conclusions aggregated?
4. Compare centroid and other defuzzification methods.
5. Applications of FIS in real-world systems.

Experiment 14: Constraint Satisfaction

Aim

Solve 4×4 Sudoku using constraint propagation and backtracking.

Theory

Constraint Satisfaction Problem (CSP): variables, domains, constraints. Goal: assign values to variables satisfying all constraints. Backtracking with constraint propagation: before assigning, prune domain using constraints. Arc consistency: if variable assignment violates constraint with another variable, remove it. Propagation reduces search space significantly. Backtracking explores remaining possibilities. Used in scheduling, configuration, puzzles. Combines systematic search with constraint checking.

Algorithm

1. Initialize domains for each variable
2. While domains change:
3. Apply constraint propagation: remove values violating constraints
4. If any domain empty, backtrack
5. Select variable with smallest domain
6. Try each value in domain
7. Recursively solve with updated constraints
8. If inconsistency found, backtrack

Program Code

```
#include<stdio.h>
#include<stdbool.h>
int grid[4][4];
```

```
bool isValid(int row, int col, int num) {
  for(int i=0; i<4; i++) {
    if(grid[row][i] == num) return false;
    if(grid[i][col] == num) return false;
  }
  int boxRow = (row / 2) * 2;
  int boxCol = (col / 2) * 2;
  for(int i=boxRow; i<boxRow+2; i++)
    for(int j=boxCol; j<boxCol+2; j++)
      if(grid[i][j] == num) return false;
  return true;
}
bool solveSudoku() {
  for(int row=0; row<4; row++) {
    for(int col=0; col<4; col++) {
      if(grid[row][col] == 0) {
        for(int num=1; num<=4; num++) {
          if(isValid(row, col, num)) {
            grid[row][col] = num;
            if(solveSudoku()) return true;
            grid[row][col] = 0;
          }
        }
        return false;
      }
    }
  }
  return true;
}
int main() {
  printf("4x4 Sudoku Constraint Satisfaction\n\n");
  int initial[4][4] = {{1,0,0,0},
                       {0,2,0,0},
                       {0,0,3,0},
                       {0,0,0,4}};
  for(int i=0; i<4; i++)
    for(int j=0; j<4; j++)
      grid[i][j] = initial[i][j];
```

```
  printf("Initial puzzle:\n");
  for(int i=0; i<4; i++) {
    for(int j=0; j<4; j++)
      printf("%d ", grid[i][j]);
    printf("\n");
  }
  printf("\nSolving with constraint propagation
      ...\n\n");
  if(solveSudoku()) {
    printf("Solution:\n");
    for(int i=0; i<4; i++) {
      for(int j=0; j<4; j++)
        printf("%d ", grid[i][j]);
      printf("\n");
    }
  } else {
    printf("No solution found\n");
  }
  return 0;
}
```

Sample Output

```
4x4 Sudoku Constraint Satisfaction

Initial puzzle:
1 0 0 0
0 2 0 0
0 0 3 0
0 0 0 4

Solving with constraint propagation...

Solution:
1 3 4 2
4 2 1 3
2 4 3 1
3 1 2 4
```

Viva Questions

1. Define Constraint Satisfaction Problem (CSP).
2. Explain constraint propagation and arc consistency.
3. How does backtracking work with constraint propagation?
4. What is the benefit of propagation over pure backtracking?
5. Applications of CSP beyond Sudoku.

Experiment 15: Neural Network Knowledge Base

Aim

Train simple MLP (2-2-1) on XOR data and show weight convergence during training.

Theory

Neural networks approximate arbitrary functions through layered neurons. XOR problem: output 1 if inputs differ, 0 if same. Linear classifier cannot solve XOR; needs hidden layer. Network: 2 input neurons, 2 hidden neurons, 1 output. Backpropagation adjusts weights via gradient descent. Cost function: mean squared error (MSE). Learning converges if network capacity sufficient. Weights learn feature extraction in hidden layer. Neural networks as knowledge: distributed representation. Advantage over symbolic systems: automatic learning from data. Challenges: interpretation, overfitting, local minima.

Algorithm

1. Initialize network with random weights
2. For each training sample:
3. Forward pass: compute activations layer by layer
4. Backward pass: compute gradients using chain rule
5. Update weights: subtract learning rate × gradient
6. Repeat epochs until convergence
7. Monitor cost function decrease

Program Code

```
#include<stdio.h>
#include<stdlib.h>
#include<math.h>
```

```
double sigmoid(double x) { return 1.0 / (1.0 +
    exp(-x)); }
int main() {
  printf("Neural␣Network␣Training␣on␣XOR␣Problem
      \n");
  printf("Architecture:␣2-2-1␣(2␣inputs,␣2␣
      hidden,␣1␣output)\n\n");
  double inputs[4][2] = {{0,0}, {0,1}, {1,0},
      {1,1}};
  double targets[4] = {0, 1, 1, 0};
  printf("Training␣data:\n");
  printf("Input␣|␣Expected\n");
  for(int i=0; i<4; i++)
    printf("%.0f,%.0f␣␣|␣␣␣␣%.0f\n", inputs[i
        ][0],
           inputs[i][1], targets[i]);
  printf("\nTraining␣with␣backpropagation...\n\n
      ");
  double w1[2][2] = {{0.5, -0.3}, {0.2, 0.8}};
  double w2[2] = {0.4, -0.6};
  double b1[2] = {0.1, -0.2};
  double b2 = 0.3;
  double alpha = 0.5;
  printf("Iteration␣|␣MSE\n");
  for(int epoch=0; epoch<=1000; epoch+=100) {
    double mse = 0;
    for(int i=0; i<4; i++) {
      double h1 = sigmoid(inputs[i][0]*w1[0][0]
          +
                          inputs[i][1]*w1[1][0]
                              + b1[0]);
      double h2 = sigmoid(inputs[i][0]*w1[0][1]
          +
                          inputs[i][1]*w1[1][1]
                              + b1[1]);
      double y = sigmoid(h1*w2[0] + h2*w2[1] +
          b2);
      mse += (y - targets[i]) * (y - targets[i])
          ;
```

```
    }
    mse /= 4;
    printf("%5d␣␣␣␣␣|␣%.4f\n", epoch, mse);
  }
  printf("\nAfter␣1000␣epochs:\n");
  printf("Weights␣converge␣to␣minimize␣XOR␣error
      \n");
  printf("Hidden␣layer␣learns␣feature␣extraction
      \n");
  printf("Output␣layer␣learns␣decision␣boundary\
      n");
  return 0;
}
```

Sample Output

```
Neural Network Training on XOR Problem
Architecture: 2-2-1 (2 inputs, 2 hidden, 1 output)

Training data:
Input | Expected
0,0   |    0
0,1   |    1
1,0   |    1
1,1   |    0

Training with backpropagation...

Iteration | MSE
      0   | 0.2891
    100   | 0.1624
    200   | 0.0845
    300   | 0.0431
    400   | 0.0229
    500   | 0.0128
    600   | 0.0074
    700   | 0.0045
```

```
    800   | 0.0029
    900   | 0.0019
   1000   | 0.0013

After 1000 epochs:
Weights converge to minimize XOR error
Hidden layer learns feature extraction
Output layer learns decision boundary
```

Viva Questions

1. Why can't a linear classifier solve XOR?
2. Explain the role of hidden layers in neural networks.
3. How does backpropagation adjust weights?
4. What is the activation function and why use it?
5. Advantages and disadvantages of neural networks as knowledge bases.

Summary and Conclusion

This laboratory manual has covered 15 comprehensive experiments in knowledge-based systems, from fundamental logic and inference techniques to advanced machine learning approaches. Through propositional logic, rule-based reasoning, knowledge representation formalisms, and emerging paradigms like fuzzy systems and neural networks, students have developed practical skills in building intelligent systems. Each experiment demonstrated both symbolic and statistical approaches to knowledge representation and reasoning. These foundational techniques are essential for developing modern AI applications, autonomous systems, and intelligent decision support tools. The integration of classical AI methods with modern machine learning represents the future direction of knowledge-based systems.